

SABER: Sensor-Agnostic Bridged Embedding Retrieval

A Unified Framework for Cross-Modal and Same-Modal
Multi-Sensor Satellite Image Retrieval

Team [your team name]

Problem Statement 11 — Bharatiya Antariksh Hackathon (BAH) 2026

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Abstract

We propose **SABER** (Sensor-Agnostic Bridged Embedding Retrieval), a unified framework for content-based retrieval of remote-sensing imagery that works *both* within a sensor modality (optical→optical, SAR→SAR, MS→MS) and *across* modalities (optical↔SAR, optical↔MS, PAN↔MS). SABER targets the three weaknesses of the current state of the art for cross-modal RS retrieval (X-JEPA): its two-encoder, two-modality design; its deterministic, instance-level latent alignment; and its brute-force floating-point nearest-neighbour search. SABER replaces these with (i) a *single* wavelength-conditioned, sensor-agnostic encoder that ingests any band configuration (SAR, multispectral, panchromatic, RGB, and — by construction — hyperspectral and elevation); (ii) a *stochastic latent bridge* based on flow matching that models the one-to-many nature of the cross-modal mapping and is distilled to a single-step predictor for fast inference; and (iii) a *relevance-geometry alignment* objective that shapes the embedding space so that distances reproduce the multi-label-overlap relevance that the evaluation actually scores, optionally in hyperbolic space to respect the land-cover hierarchy. Retrieval is made sub-millisecond through end-to-end learned compact codes indexed with FAISS and refined by an uncertainty-aware graph re-ranking step. The design is engineered to win on the two criteria that carry the most weight in the evaluation — cross-modal F_1 and average retrieval time per query — while remaining accurate on same-modal and single-label benchmarks.

1 Problem statement and objective

Earth-observation archives are growing faster than they can be searched. The same geographic scene is captured by heterogeneous sensors — optical/RGB, multispectral (MS), hyperspectral, Synthetic Aperture Radar (SAR), and elevation — each recording different physics. Metadata search fails when semantically similar land cover appears under different seasons, conditions, or *sensors*. The task (Problem Statement 11) is to build a retrieval system that, given a query image from one modality, returns a ranked list of semantically relevant gallery images from the *same* or a *different* modality, learning a common representation space in which scenes with similar land-cover / land-use are close regardless of sensor.

Requirements. (1) Same-modal retrieval; (2) cross-modal retrieval (weighted more heavily, being harder); (3) top-5 and top-10 ranking; and (4) *low average retrieval time per query*. The system is scored by $F_1@5$ and $F_1@10$ for same- and cross-modal retrieval, by precision and mean average precision for single-label benchmarks, and by retrieval latency.

2 Datasets and evaluation protocol

We adopt the standard dual-modality retrieval benchmarks (Table 1) and the four retrieval directions $a \rightarrow a$, $b \rightarrow b$ (same-modal) and $a \rightarrow b$, $b \rightarrow a$ (cross-modal); trivial self-matches are excluded.

Table 1: Benchmarks, modalities, relevance definitions, and reported metrics.

Benchmark	Modalities	Relevance	Metric
BEN-14K	S1 SAR (2 ch.) $\leftrightarrow$ S2 MS (12 ch.)	multi-label overlap (19 cls.)	$F_1@5$, $F_1@10$
DSRSID	Gaofen PAN (1 ch.) $\leftrightarrow$ MS (4 ch.)	single-label class equality (8)	$P@5$, mAP
CBRSIR_VS	PAN $\leftrightarrow$ MS (single-label)	class equality	$P@5$, mAP

Multi-label relevance ($F_1@K$). For a query q and a retrieved item r with label sets $\mathcal{Y}_q, \mathcal{Y}_r$, precision, recall, and F_1 are $P(q, r) = \frac{|\mathcal{Y}_q \cap \mathcal{Y}_r|}{|\mathcal{Y}_r|}$, $R(q, r) = \frac{|\mathcal{Y}_q \cap \mathcal{Y}_r|}{|\mathcal{Y}_q|}$, and $F_1(q, r) = \frac{2PR}{P+R+\varepsilon}$; the final $F_1@K$ averages item-level F_1 over the top- K and then over all queries. **Single-label relevance.** A retrieved image is relevant iff it shares the query’s class; we report $P@5$ and global mAP.

3 Background and the gap we exploit

The intended lineage is **CSMAE** $\rightarrow$ **REJEPa** $\rightarrow$ **X-JEPa**. CSMAE [3] adapts masked autoencoders to cross-sensor CBIR with intra-/inter-modal reconstruction. REJEPa [2] is the first joint-embedding predictive architecture (JEPa) for *unimodal* RS retrieval: it predicts target-token features from context tokens (no pixel reconstruction, no negatives) with VICReg regularization, cutting compute by 40–60%. **X-JEPa** [1] is the current cross-modal SOTA: dual ViT-B/16 encoders, a cross-attention predictor with shared learnable queries, a geometry-aware *Prediction-Space-Alignment* (PSA, Mahalanobis) loss, and VICReg; retrieval uses floating-point k -NN.

Three exploitable weaknesses (Table 2) motivate SABER: the architecture is hard-wired to *two* modalities through *separate* encoders trained largely from scratch on $\sim 14K$ pairs; the alignment is *deterministic and instance-level*, mismatched to a multi-label-overlap metric; and retrieval is *brute-force float k -NN*, ignoring the scored latency criterion.

Table 2: Weaknesses of the incumbent (X-JEPa / CSMAE / REJEPa) and SABER’s response.

Gap in the incumbent	SABER’s response
Two separate encoders, exactly two modalities, trained from scratch on a small set.	One <i>wavelength-conditioned</i> backbone for N modalities, warm-started from a remote-sensing foundation model.
Deterministic, instance-level alignment (PSA + VICReg) mismatched to the metric.	<i>Stochastic latent bridge</i> (one-to-many) + <i>relevance-geometry</i> alignment matched to multi-label overlap.
Brute-force floating-point k -NN search; latency ignored.	End-to-end <i>compact codes</i> + FAISS ANN + graph re-ranking $\Rightarrow$ sub-millisecond queries.

4 Proposed method: SABER

4.1 Overview

SABER learns a single shared embedding space $\mathcal{Z} \subset \mathbb{R}^d$ into which images of *any* modality are mapped by one encoder, such that semantic neighbours are close irrespective of sensor. At *training* time a stochastic latent bridge teaches the space to be modality-invariant by predicting the latent of one modality from another, and a relevance-geometry loss shapes distances to match the evaluation metric. At *inference* time the heavy machinery is gone: an image is encoded once, hashed to a compact code, and matched against a pre-indexed gallery with FAISS, followed by a

cheap graph re-ranking of the top- K . Figure 1 shows the inference path; Figure 4 shows the training architecture.

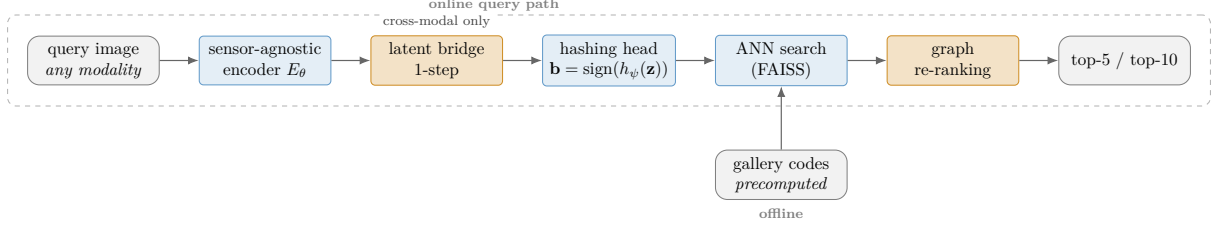


Figure 1: SABER inference pipeline. The query is encoded once; for cross-modal queries an optional single-step latent bridge translates it toward the target-modality manifold. Compact codes are matched in a FAISS index built over precomputed gallery codes, and the top- K shortlist is re-ranked. All expensive computation lives in training; inference is a single forward pass plus an ANN lookup.

4.2 Module 1 — Sensor-agnostic encoder

Instead of one encoder per modality, SABER uses a *single* Vision-Transformer backbone with a *wavelength-conditioned dynamic patch embedding* (Figure 2), following the DOFA design [5]. A hypernetwork g_ω takes the central wavelength λ_c of each input channel and *generates* the patch-projection weights for that channel:

$$W_c = g_\omega(\lambda_c), \quad \mathbf{p}_n = \sum_c W_c x_{n,c} + \mathbf{e}_{\text{pos}}(n), \quad (1)$$

so a 2-band SAR patch, a 12-band Sentinel-2 patch, a 1-band Gaofen panchromatic patch, and a 4-band Gaofen MS patch all produce tokens for the *same* transformer E_θ . This (a) realises the “common space irrespective of sensor” requirement at the input level, (b) extends for free to hyperspectral (EnMAP) and elevation (DEM) channels, and (c) lets us warm-start from a foundation model pretrained on millions of multi-sensor images (DOFA [5], Galileo [6], or TerraMind [7]), which is strictly stronger than X-JEPA’s from-scratch ViT on 14K pairs. We keep the backbone frozen and adapt $\sim 1\text{--}2\%$ of parameters with LoRA, making training fast and data-efficient.

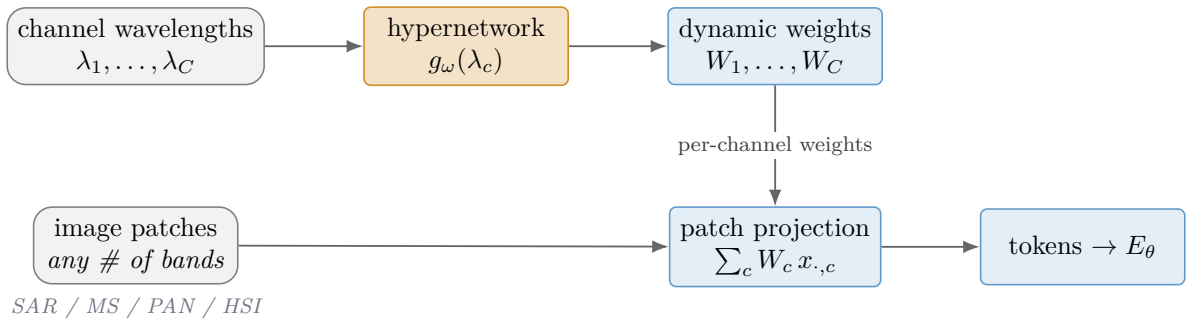


Figure 2: Wavelength-conditioned dynamic patch embedding (Module 1). The hypernetwork generates per-channel projection weights from wavelengths, so one backbone consumes arbitrary sensors and band counts.

4.3 Module 2 — Stochastic latent bridge

Cross-modal correspondence is fundamentally *one-to-many*: a SAR texture is consistent with many optical appearances and vice versa. X-JEPA’s deterministic predictor collapses this to

a single point. SABER instead learns the *conditional distribution* $p(\mathbf{z}_b \mid \mathbf{z}_a)$ of the target-modality latent given the source-modality context, using a *flow-matching* (rectified-flow / Schrödinger-bridge) predictor in latent space [9, 10]. Let $\mathbf{z}_a = E_\theta(x_a)$ (context, source modality) and $\mathbf{z}_b = E'_\theta(x_b)$ (target, EMA encoder, stop-gradient). Define the linear interpolant $\mathbf{z}_\tau = (1 - \tau)\mathbf{z}_a + \tau\mathbf{z}_b + \sigma\epsilon$, $\tau \sim \mathcal{U}[0, 1]$, $\epsilon \sim \mathcal{N}(0, I)$. A conditional velocity field v_ϕ is trained with the conditional flow-matching loss

$$\mathcal{L}_{\text{bridge}} = \mathbb{E}_{\tau, \epsilon} \|v_\phi(\mathbf{z}_\tau, \tau, c_a, \mathbf{s}) - (\mathbf{z}_b - \mathbf{z}_a)\|_2^2, \quad (2)$$

where c_a is the source context and $\mathbf{s}$ are shared learnable queries that act as modality-agnostic semantic anchors. Integrating v_ϕ from $\mathbf{z}_a$ yields the predicted target latent $\hat{\mathbf{z}}_{a \rightarrow b} = \mathbf{z}_a + \int_0^1 v_\phi d\tau$. The map is trained symmetrically ($a \rightarrow b$ and $b \rightarrow a$).

Fast inference by distillation. A multi-step ODE solve would be slow, so we *distil* v_ϕ into a single-step predictor P_ϕ (mean-flow / consistency distillation [11]): $\hat{\mathbf{z}}_{a \rightarrow b} \approx P_\phi(\mathbf{z}_a, c_a, \mathbf{s})$, one forward pass. The bridge’s residual variance gives a per-query *uncertainty* $u(q)$ that we feed to re-ranking (Module 4). Figure 3 illustrates the bridge.

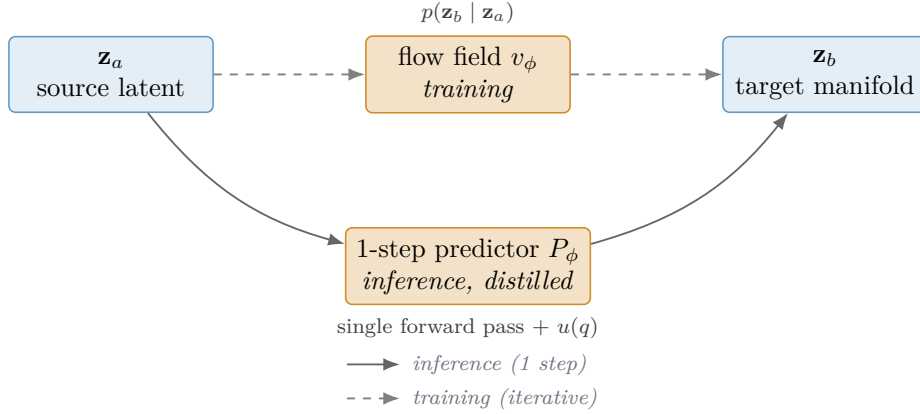


Figure 3: Stochastic latent bridge (Module 2). A flow-matching field learns the one-to-many cross-modal mapping in latent space (dashed, training) and is distilled to a single-step predictor (solid, inference) that also exposes a calibrated uncertainty for re-ranking.

4.4 Module 3 — Relevance-geometry alignment

The metric rewards *multi-label overlap*, not instance identity. SABER therefore shapes the space so that embedding similarity reproduces label overlap. For a pair (i, j) with label sets $\mathcal{Y}_i, \mathcal{Y}_j$, define the soft relevance target $s_{ij} = \frac{|\mathcal{Y}_i \cap \mathcal{Y}_j|}{|\mathcal{Y}_i \cup \mathcal{Y}_j|}$ (Jaccard). The alignment loss regresses cosine similarity onto s_{ij} with a listwise ranking term $\mathcal{L}_{\text{rank}}$ that preserves the relevance order in the top neighbourhood:

$$\mathcal{L}_{\text{rel}} = \mathbb{E}_{i,j} (\cos(\mathbf{z}_i, \mathbf{z}_j) - s_{ij})^2 + \beta \mathcal{L}_{\text{rank}}. \quad (3)$$

For single-label benchmarks (DSRSID, CBR SIR_VS) this degenerates to $s_{ij} \in \{0, 1\}$ (class equality), directly optimising $P@5/\text{mAP}$. Because land-cover labels form a hierarchy (CORINE Level-3), we offer a *hyperbolic* variant: embed $\mathbf{z}$ in the Poincaré ball and match the hyperbolic distance $d_{\mathbb{H}}$ to $(1 - s_{ij})$, which represents tree-like label structure with low distortion [12]. A VICReg term $\mathcal{L}_{\text{vic}}$ (variance, invariance, covariance) [8] prevents representational collapse, as in REJEP A/X-JEP A.

4.5 Module 4 — Compact codes, ANN index, and re-ranking

To win the latency criterion we co-train a hashing head h_ψ that maps embeddings to compact codes $\mathbf{b} = \text{sign}(h_\psi(\mathbf{z}))$ (relaxed with tanh during training), with a similarity-preserving hashing loss and a quantisation penalty:

$$\mathcal{L}_{\text{hash}} = \mathbb{E}_{i,j} \left(\frac{1}{m} h_\psi(\mathbf{z}_i)^\top h_\psi(\mathbf{z}_j) - s_{ij} \right)^2 + \gamma \mathbb{E}_i \| |h_\psi(\mathbf{z}_i)| - \mathbf{1} \|_2^2. \quad (4)$$

Codes are indexed with FAISS (4-bit product quantisation with FastScan, or a binary index for Hamming search), giving order-of-magnitude speedups over brute-force float k -NN [13]. A shortlist of the top- K ($K \leq 100$) is re-ranked by k -reciprocal neighbours and label co-occurrence (a modern, differentiable echo of the classical semisupervised graph-theoretic retrieval idea [14]), weighted by the bridge uncertainty $u(q)$. The re-rank is vectorised over a tiny candidate set and adds negligible latency while lifting $F_1 @5/@10$, especially cross-modal.

4.6 Training objective

The encoder (LoRA), bridge, alignment, and hashing heads are trained jointly:

$$\mathcal{L}_{\text{SABER}} = \mathcal{L}_{\text{bridge}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}} + \lambda_{\text{vic}} \mathcal{L}_{\text{vic}} + \lambda_{\text{hash}} \mathcal{L}_{\text{hash}}. \quad (5)$$

The target encoder E'_θ is the exponential moving average of E_θ (stop-gradient on targets), preventing collapse without negatives. Figure 4 shows the full training graph.

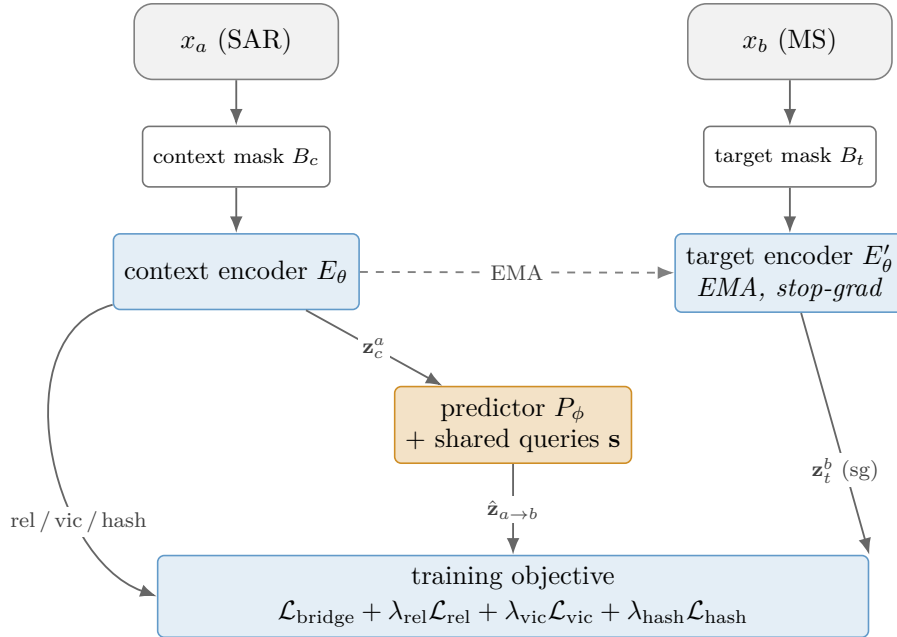


Figure 4: SABER training architecture (shown for the $a \rightarrow b$ direction; the symmetric $b \rightarrow a$ pass is omitted for clarity). A shared sensor-agnostic encoder produces the context latent $\mathbf{z}_c^a$; an EMA target encoder produces the stop-gradient target $\mathbf{z}_t^b$. The predictor, conditioned on shared learnable queries, forecasts $\hat{\mathbf{z}}_{a \rightarrow b}$. The four terms — flow-matching bridge, relevance-geometry, VICReg, and hashing — form one joint objective; the predictor and EMA target encoder are not used at inference.

5 Why SABER is simultaneously fast and accurate

A common objection is that generative bridges and re-ranking are slow. SABER resolves this by pushing all expensive computation into *training* (Table 3). Inference is one encoder forward

pass, one distilled bridge step (cross-modal only), a compact-code ANN lookup, and a re-rank over ≤ 100 candidates.

Table 3: Where latency could appear, and how SABER removes it.

Potential cost	SABER’s mitigation
Generative bridge at query time	Distilled to a single forward step
Float brute-force search	Product-quantised / binary codes + FAISS
From-scratch backbone training	Frozen foundation model + LoRA
Re-ranking overhead	Restricted to the top- K shortlist, vectorised

6 Novelty and contributions

1. **First sensor-agnostic encoder for cross-modal RS retrieval:** a single wavelength-conditioned backbone for N modalities, replacing the two-encoder, two-modality design and extensible to hyperspectral and elevation.
2. **First stochastic latent bridge for RS-CMIR:** a flow-matching predictor that models the one-to-many cross-modal mapping and is distilled to a single-step, uncertainty-aware predictor for fast inference.
3. **Relevance-geometry alignment matched to the metric:** the embedding geometry (optionally hyperbolic) is trained so distances reproduce multi-label overlap, rather than instance identity.
4. **First JEPA-lineage retriever with end-to-end compact codes:** learned hashing + FAISS + uncertainty-weighted graph re-ranking, directly optimising the average-retrieval-time criterion.

7 Experimental plan

Baselines. CSMAE-SESD [3], REJEPA [2], X-JEPA [1], and frozen foundation-model k -NN (DOFA [5], Galileo [6]). **Metrics.** $F_1@5, F_1@10$ for all four directions on BEN-14K; $P@5$, mAP on DRSID and CBRISIR_VS; average retrieval time per query; parameters and GFLOPs. **Ablations.** bridge (stochastic) vs. deterministic predictor; relevance-geometry vs. PSA; Euclidean vs. hyperbolic; float vs. PQ/binary codes (accuracy–latency curve); re-ranking on/off; backbone choice. Table 4 states the SOTA to beat and SABER’s targets.

Table 4: BEN-14K $F_1@5$ (%): incumbent SOTA (X-JEPA) and SABER targets. Cross-modal directions, weighted most heavily, are the primary objective. SABER figures are projected targets, not yet measured.

Direction	X-JEPA (SOTA)	SABER (target)
S1→S1 (SAR, same-modal)	72.98	≥ 75
S2→S2 (MS, same-modal)	82.65	≥ 84
S1→S2 (cross-modal)	61.23	$\geq$ 66
S2→S1 (cross-modal)	63.73	$\geq$ 68
Params / search	172.9 M / float k -NN	smaller / sub-ms ANN

8 Technology stack and engineering blueprint

SABER is built entirely from open, pip-installable components with publicly downloadable model weights — no proprietary dependencies and nothing that must be trained from scratch before we see results. Table 5 summarises the stack layer by layer; the subsections give the concrete choices, load commands, and hardware budget.

Table 5: End-to-end technology stack. Every component is open-source and the foundation-model weights are publicly hosted (Hugging Face / Torch Hub).

Layer	Tooling	Role in SABER
Backbone (FM)	DOFA ViT-B/16 (Torch Hub / Torch-Geo); alt. Galileo (nasaharvest), TerraMind (TerraTorch) [5, 6, 7]	Sensor-agnostic encoder, 768-D embeddings; wavelength-conditioned patch embed
Adaptation	Hugging Face PEFT (LoRA) [19]	Train $\sim 1\text{--}2\%$ of backbone params; fast, data-efficient
Cross-modal bridge	torchcfm / flow_matching [20]	Conditional flow-matching predictor + single-step distillation
Geometry	geoopt [21] (Poincaré ball, Riemannian Adam)	Optional hyperbolic relevance-geometry embedding
Data IO	TorchGeo datasets/datamodules [17], rasterio , rioxarray , NumPy	BigEarthNet-MM loaders; custom DSR-SID / CBR SIR_VS reader
Augmentation	Kornia / Albumentations	GPU, modality-specific transforms (no RGB-only ops)
Training	PyTorch 2.x + Lightning [22], bf16 AMP, torch.compile ; Hydra configs; Weights & Biases	Mixed-precision training, sweeps, experiment tracking
Retrieval	FAISS (faiss-gpu) [13]	IVF-PQ FastScan / binary HNSW index; uncertainty-weighted graph re-rank
Serving / demo	FastAPI + Gradio ; TorchScript/ONNX export; Docker	Query API + interactive top-5/10 viewer with timing
Hardware	1× NVIDIA A100 40 GB (or RTX 4090 24 GB / Colab / Kaggle), CUDA 12.x	Single-GPU training; CPU-only inference is viable for the gallery sizes here

8.1 Backbone and modality handling

The encoder is the one place modality-agnosticism must be solved, and DOFA solves it at the input: a hypernetwork turns per-channel wavelengths into patch-projection weights, so SAR (2 bands), Sentinel-2 (12), Gaofen PAN (1), and Gaofen MS (4) all map into the *same* 768-D space with one set of weights. Loading is a single call.

```
import torch
# Primary sensor-agnostic backbone: DOFA ViT-B/16 (768-D output)
model = torch.hub.load("zhu-xlab/DOFA", "vit_base_dofa", pretrained=True)
# Per-channel central wavelengths (micrometres); same call for S1/S2/PAN/MS/HSI
wvs = [0.49, 0.56, 0.665, 0.842, 1.61, 2.19] # example band subset
z = model.forward_features(x, wvs) # x: [B, C, 224, 224] -> [B, 768]
# Alternative loaders / backbones:
# TorchGeo : dof_base_patch16_224(weights=DOFABase16_Weights.DOFA_MAE)
# Galileo : hf download nasaharvest/galileo --include "models/**"
# TerraMind: BACKBONE_REGISTRY.build('terramind_v1_base', pretrained=True,
#                                     modalities=['S2L2A', 'S1GRD'])
```


8.2 Data engineering

Sentinel-1/2 pairs come straight from TorchGeo’s BigEarthNet datamodule; the 19-class nomenclature and the BEN-14K split (Serbia, summer) are applied on top. DSRSID (Gaofen PAN/MS) and CBR SIR_VS use a thin `rasterio`-based reader. Each modality is normalised independently (S1 in dB; S2/MS per-band mean-std; PAN min-max), then resized to the backbone’s 224×224 input.

```
from torchgeo.datasets import BigEarthNet
ds = BigEarthNet(root="data", split="train", bands="all",
                 num_classes=19, download=True) # 12 S2 + 2 S1 bands, 19 labels
```

Pretraining data. Self-supervised alignment is trained on BigEarthNet-MM (590K S1/S2 pairs) [4], optionally extended with SSL4EO-S12; the land-cover labels supervise only the relevance head (semi-supervised). The bridge needs no labels.

8.3 Cross-modal bridge, geometry, and training

The flow-matching bridge (Section 4.3) is implemented with `torchcfm`; the optional hyperbolic head (Section 4.4) uses `geoopt`’s Poincaré ball with Riemannian Adam. Everything is wrapped in a single Lightning module with bf16 mixed precision and `torch.compile`; configs are managed by Hydra and runs logged to Weights & Biases. Because the backbone is frozen (LoRA-adapted), only $\sim 10\text{--}30$ M parameters are trainable.

8.4 Retrieval and indexing

Gallery embeddings/codes are computed once offline and indexed in FAISS; queries hit the index online. Table 6 lists the index options and the accuracy–latency trade-off; we report the full curve and pick the operating point that maximises score under the latency criterion.

Table 6: FAISS index options and the accuracy–latency trade-off SABER exploits.

Index	Code size	Speed	Use
IndexFlatIP (float)	768×32 bit	exact, slow	accuracy ceiling / baseline
IVFPQ FastScan (4-bit)	64×4 bit	$\sim 1\text{M}$ QPS	primary float-code index
IndexBinaryHNSW (Hamming)	256 bit	$\mu\text{s}/\text{query}$	learned binary codes
HNSWFlat (float)	768×32 bit	fast, exact-ish	high-RAM accurate fallback

```
import faiss
d = 768
# Compressed ANN: inverted file + 4-bit PQ (FastScan) -> ~1e6 QPS
index = faiss.IndexIVFPQ(faiss.IndexFlatIP(d), d, 1024, 64, 4)
index.train(gallery); index.add(gallery); index.nprobe = 16
D, I = index.search(query, 10) # top-10 neighbours
# Learned binary codes -> Hamming search, microsecond latency
bidx = faiss.IndexBinaryHNSW(256, 32)
bidx.add(gallery_codes); Db, Ib = bidx.search(query_codes, 10)
```

8.5 Serving and demo

Figure 5 shows the deployment: an *offline* lane builds the FAISS index over the gallery, and an *online* lane serves queries through the encoder, single-step bridge (cross-modal only), hashing,

ANN search, and re-rank, exposed via a FastAPI endpoint and a Gradio UI that shows the ranked results and the measured retrieval time per query. The trained artifact (LoRA + bridge + hashing weights) is shared by both lanes.

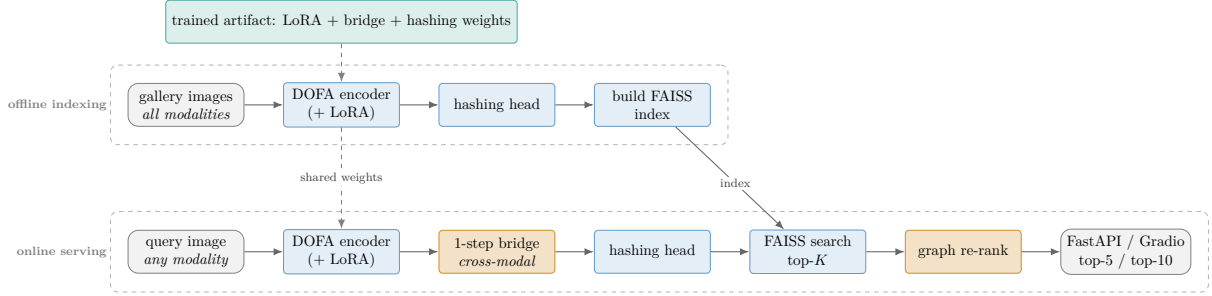


Figure 5: Deployment architecture. Offline gallery indexing (top) and online query serving (bottom) share one trained artifact. Heavy work (encoding, index build) is amortised offline; the online path is a single forward pass plus a sub-millisecond ANN lookup and a top- K re-rank.

8.6 Hardware budget and reproducibility

With a frozen, LoRA-adapted backbone and bf16 AMP, alignment pretraining on BigEarthNet-MM fits comfortably on a *single* A100 40 GB (batch 256–512, gradient checkpointing if needed) in on the order of a day; per-benchmark fine-tuning takes hours. Gallery encoding is a one-off offline pass. At inference the encoder runs in ~ 10 – 20 ms/image on GPU (and is CPU-feasible at these gallery sizes), while ANN search is sub-millisecond. For reproducibility we ship Hydra configs, fixed seeds, W&B run logs, a Dockerfile, model cards, and the released training/eval code.

9 Roadmap and risk mitigation

Phased plan.

Phase 0. Data pipelines + frozen-FM k -NN baselines; reproduce X-JEPA/CSMAE numbers.

Phase 1. Module 1 + Euclidean relevance-geometry $\Rightarrow$ target same-modal SOTA.

Phase 2. Module 2 latent bridge + distillation $\Rightarrow$ target cross-modal SOTA.

Phase 3. Hyperbolic variant + hashing/FAISS + re-ranking $\Rightarrow$ latency + final F_1 lift.

Phase 4. Optional pixel-space modality completion; full ablations; demo UI.

Risk mitigation. Each phase degrades gracefully to the previous working state: if the flow bridge underperforms we fall back to a deterministic predictor; if the hyperbolic head is unstable we ship the Euclidean relevance-geometry (already novel and metric-aligned); foundation-model domain gaps on Gaofen PAN are absorbed by wavelength conditioning or a light fine-tune. Every dependency has a drop-in substitute (DOFA $\rightarrow$ Galileo/TerraMind; FAISS $\rightarrow$ ScaNN; Gradio $\rightarrow$ Streamlit).

10 Conclusion

SABER keeps what makes the JEPA lineage strong — latent prediction, no negatives, collapse-free training — and removes its three weak spots with a sensor-agnostic backbone, a probabilistic

latent bridge, and a metric-aligned, indexed retrieval stack. It is engineered to win the two highest-weighted criteria — cross-modal F_1 and average retrieval time per query — while remaining accurate on same-modal and single-label tasks, and every component constitutes a clean, defensible contribution.

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